

Loops Lab Extended Abstract

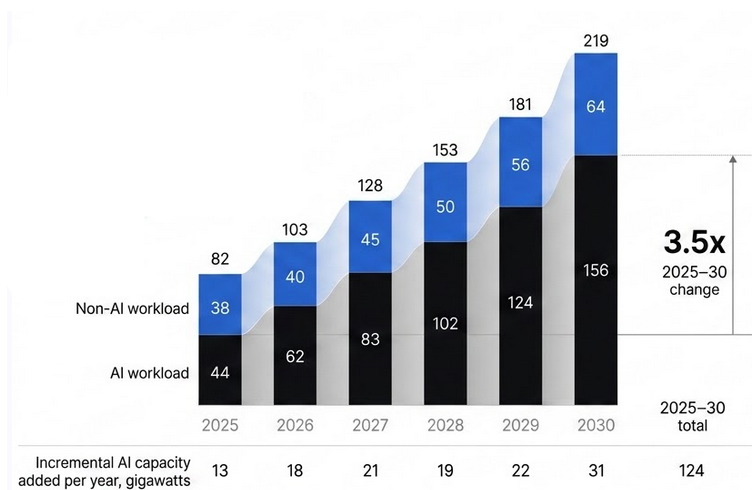
Executive Summary

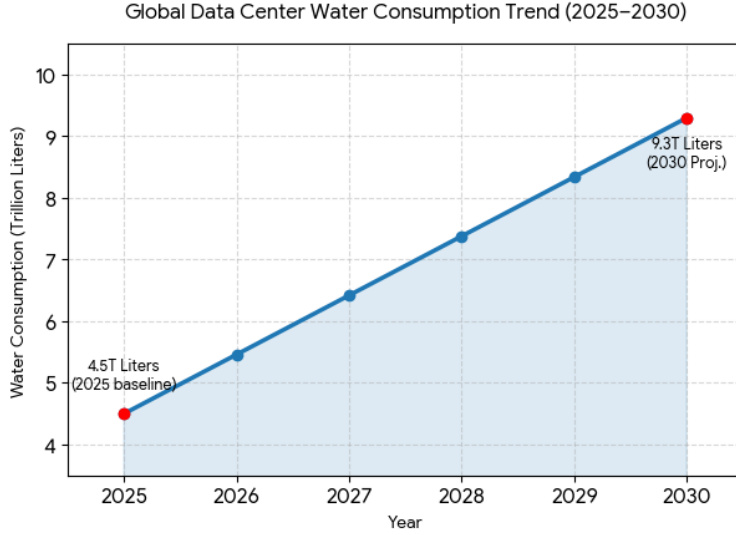
The exponential growth of Artificial Intelligence—driven by massive Large Language Models (LLMs)—is actively colliding with the hard limits of standard electronic silicon. As computational demands for AI usage skyrocket, datacenters are crippled by unsustainable energy consumption, extreme thermal dissipation, and water drain.

What we need is a disruptive paradigm shift **Optical AI Accelerators leveraging Thin-Film Lithium Niobate (LiNbO_3)**. Computing at the speed of light, our architecture slashes energy costs by 90% and makes dissipative cooling systems obsolete. Overcoming historical scalability bottlenecks, we introduce a novel **loop architecture** and native **optical nonlinearities**. The result is a high-fidelity, ultra-low-power processing core capable of liberating large AI models from high-performance computing (HPC) infrastructures and deploying them seamlessly to the Edge.

1 The AI Computational Bottleneck

Modern deep learning relies on trillions of floating-point operations. While multi-core CPUs and many-core GPUs have historically provided the necessary parallelization [2], they have hit the “power wall.” Traditional CMOS processors are bottlenecked by extreme power requirements and severe thermal output, with an associated energy cost up to 10 to 100 pJ per operation, requiring billions of liters of water for cooling systems [3], making further scaling computationally and financially unsustainable.





2 The Lithium Niobate (LiNbO_3) Advantage

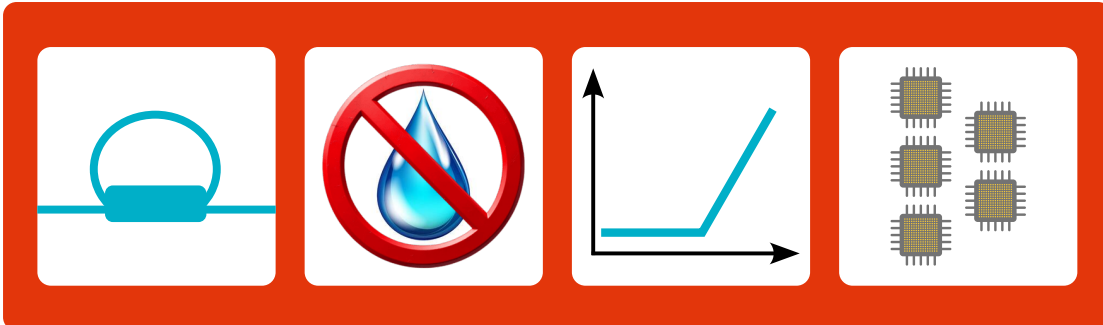
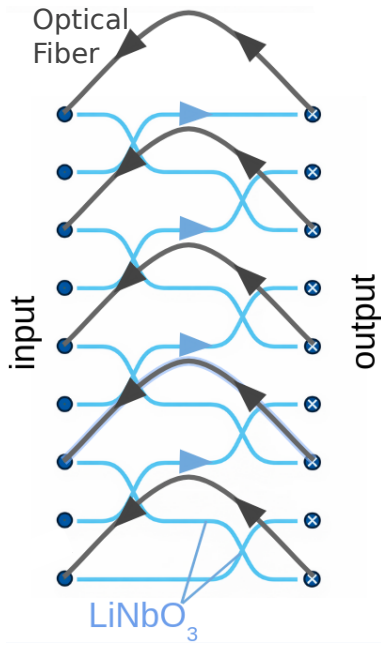
Optical computing fundamentally shifts this paradigm. Contrary to digital hardware, on optical devices AI operations can be performed naturally with little amount of energy. However, previous silicon-based optical chips struggled with analog noise due to the leakage of optical losses, lack of native non-linearity, slow reprogrammability, high static energy consumption due to the dissipative thermo-optic technology and power-hungry optical-electrical-optical conversions.

Thin-film Lithium Niobate (TFLN), which has recently driven particular attention in the alternative field of quantum photonic computing, is a novel technology envisaged to solve these critical hurdles. With respect to silicon technologies for classical optical computing, TFLN allows high-speed reprogrammability, with low optical losses compared to silicon and virtually zero static energy consumption [1]. Tuning LiNbO_3 modulators requires a mere 1 – 10 pJ per operation, avoiding static power consumption and rendering computational energy costs negligible compared to their digital counterparts.

3 Core Architectural Innovations

To transcend standard “proof of concept” models and achieve deployable LLM inference, our joint work introduces three fundamental design breakthroughs:

- **Loop Architecture:** To reduce the physical footprint of the chip and its scalability, we rely on time-bin encoding. A spatial mesh is connected via optical fiber loops, recycling hardware components for successive layers.
- **Native Optical activation functions:** Traditional optical chips route data back to electronics for non-linear activation, inhibiting the energy advantage of the optical approach. We introduce a purely optical nonlinear activation function based on standard optical techniques (homodyne detection).
- **Modularity:** Addressing the channel limit of current processors, we employ proper modularization techniques in linear algebra to combine multiple loop-based processing units, successfully scaling beyond such a channel ceiling.



4 Empirical Performance Comparison

The innovations described in the previous section are backed by the characteristic advantages of optical computing: dumped energy consumptions and no cooling systems required, which e.g. reduces the demand of water supply in datacenters. Table 1 summarizes the technological leap from traditional electronic and legacy silicon architectures to the proposed LiNbO₃ framework.

Metric	Traditional CMOS (GPU)	Proposed LiNbO ₃ Platform
Energy per Operation	10 – 100 pJ	1 – 10 pJ
Thermal Dissipation	High (1.5x PUE ¹)	Near Zero ($\sim$ 1.0x PUE)
Non-Linearity (Activation)	Native (Digital)	Native (Optical)
Fidelity	Exact (Digital)	Ultra-High (98.5%) [1]

Table 1: Architectural benchmark comparing the limitations of current standards against the proposed TFLN design.

5 Development Roadmap & Conclusion

Our path to ubiquitous optical AI progresses through three strategic milestones: 1) **Foundational Models** (FNNs/MLPs) targeting standard algebraic workloads; 2) **Spatial/Vision Processing** (CNNs/ResNets) executing massively parallel localized convolutions; and ultimately, 3) **Large Language Models** (Transformers) performing multi-trillion parameter attention mechanisms natively.

The Future of Ubiquitous AI

By dismantling the power grid bottleneck, this architecture will pull LLMs out of hyperscale clouds and securely onto Edge devices—from autonomous robotics to consumer electronics. We are not just making datacenters greener; we are democratizing artificial intelligence.

References

- [1] Y. Zheng, R. Wu, et al. “Photonic Neural Network Fabricated on Thin Film Lithium Niobate for High-Fidelity and Power-Efficient Matrix Computation.” *arXiv preprint arXiv:2402.16513* (2024).
- [2] M. Q. Ansari and M. Q. Ansari. “Accelerating Matrix Multiplication: A Performance Comparison Between Multi-Core CPU and GPU.” *arXiv preprint arXiv:2507.19723* (2025).
- [3] Z. Barnett-Itzhaki, “The water footprint of artificial intelligence: Emerging solutions and governance imperatives.” *Water Research* (2026).